DO NOW — RECALL

Arrays, records and tuples

Name: _____ Class: _____ Date: _____

1. Why must exponents match before adding binary mantissas?

2. What does passing an int by value mean?

3. What is values[1] for int[] values={4,6,9}?

4. a={1,2,3}; a[0]=a[1]+a[2]. Find a.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The mantissa bit positions must have matching place values.

2. Model answer

The method receives a copy of the integer.

3. Model answer

6.

4. Model answer

[5,2,3].

Vocabulary — know these terms

array — A fixed-length indexed collection of values of one element type in C#.

index — Zero-based position of an array element.

record — A structure that groups related named fields or properties.

tuple — A compact ordered grouping of multiple values.

assignment — Store a new value at a variable or array location.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm and explain

Put 6 into index 2: [3,9,6].

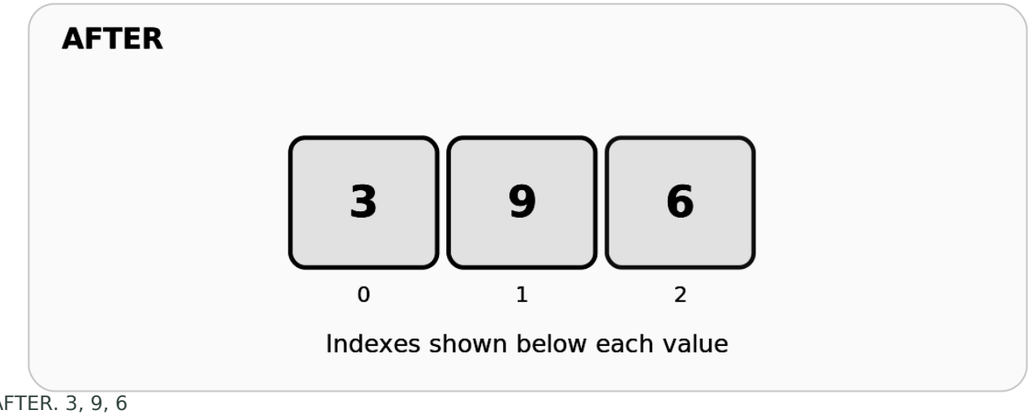
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a = {3,9,5}; a[2] = a[0] * 2.

a[0] is 3, so 3 × 2 = 6.



AFTER. 3, 9, 6

NEW LEARNING

Read, annotate and explain

Zero-based indexing

scores[0] is 4 and scores[2] is 2.

Read before writing

The right-hand expression is 4 + 2 = 6.

Replace the selected value

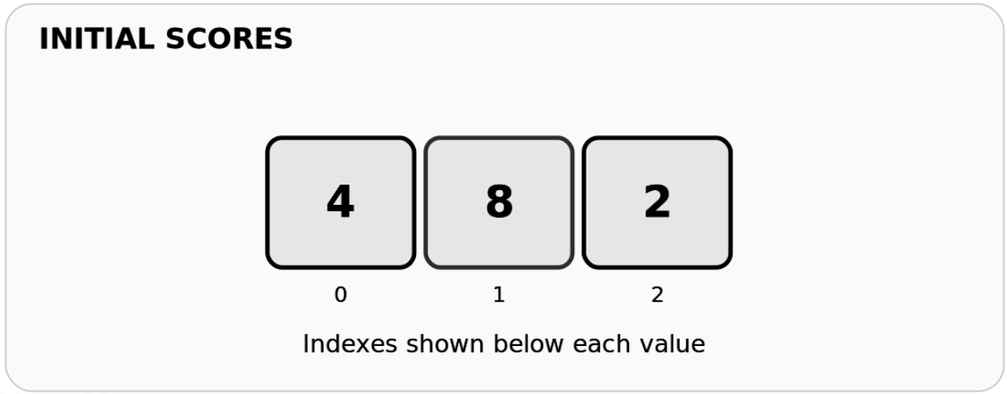
Assign 6 at index 1; the array now contains 4, 6, 2.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

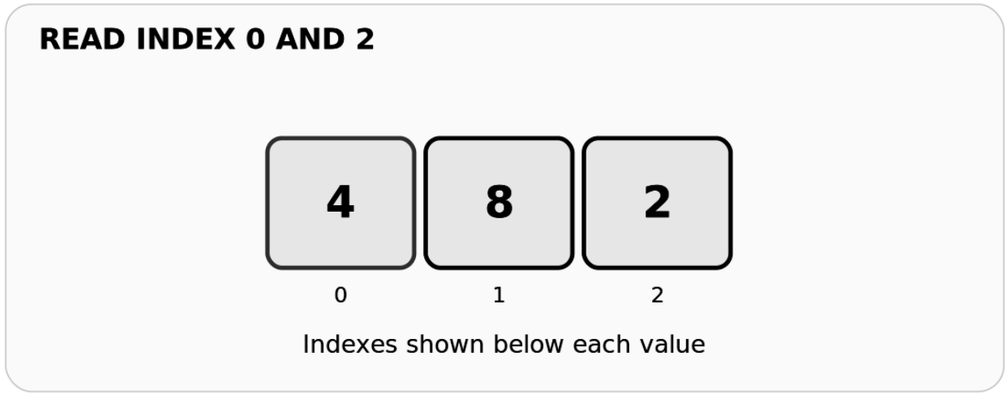
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INITIAL SCORES. 4, 8, 2

New Learning



READ INDEX 0 AND 2. 4, 8, 2

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An array with three values has a last index of 3.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

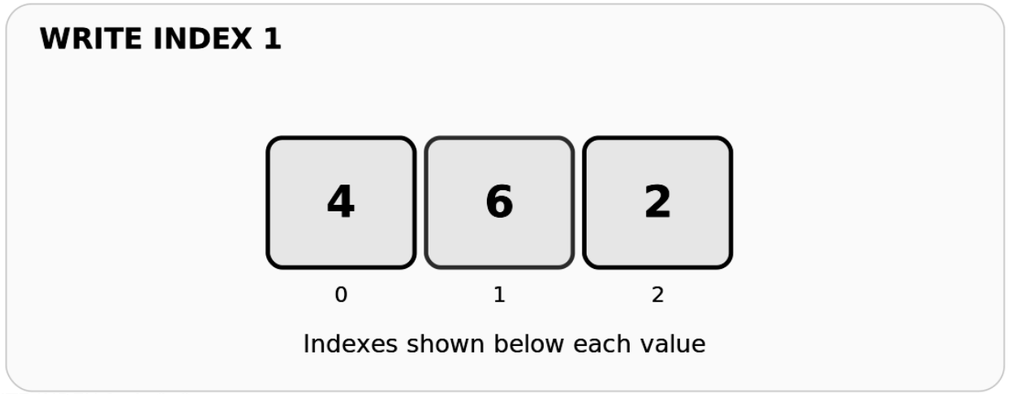
Question 3

Explain a suitable use of a named tuple in C# and one limitation.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



WRITE INDEX 1. 4, 6, 2

I DO — WORKED EXAMPLE

Follow the worked method

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

Predict

marks[0] + marks[1] means 5 + 7.

Confirm and explain

12 replaces the value at index 2: [5,7,12].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give the final array after int[] a={8,4,2}; a[1]=a[0]-a[2];

Question 2

When would you choose a record rather than a simple array for a pupil's name and mark?

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict a[0] is 3, so 3 × 2 = 6.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

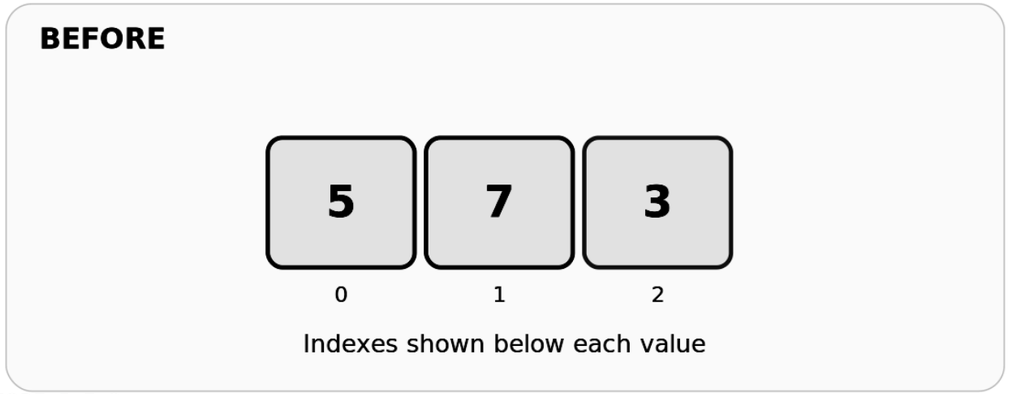
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

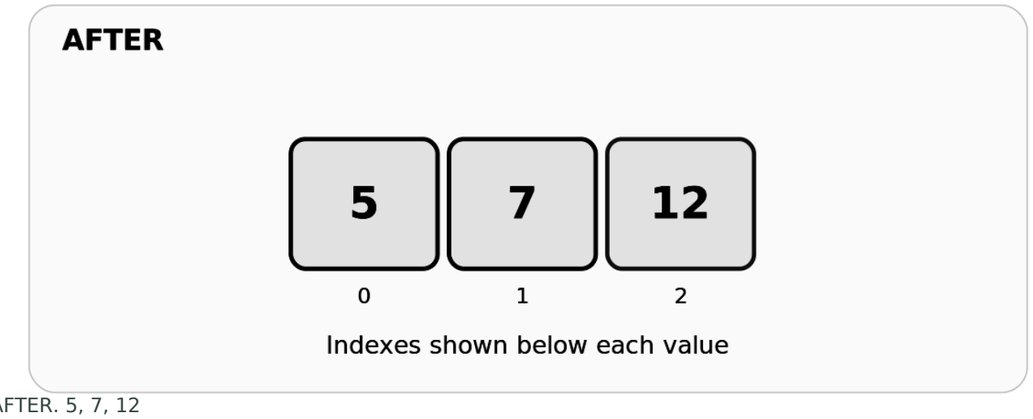
marks[0] + marks[1] means 5 + 7.



BEFORE. 5, 7, 3

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

marks[0] + marks[1] means 5 + 7.



AFTER. 5, 7, 12

WE DO — GUIDED PRACTICE

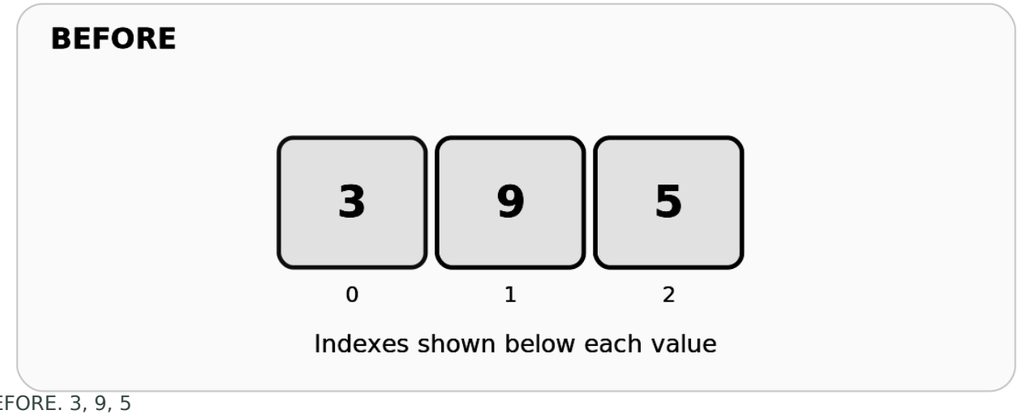
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a = {3,9,5}; a[2] = a[0] * 2.

a[0] is 3, so 3 × 2 = 6.



BEFORE. 3, 9, 5